

Add-on Occlusion: An External Module for Optical See-through Augmented Reality Displays to Support Mutual Occlusion

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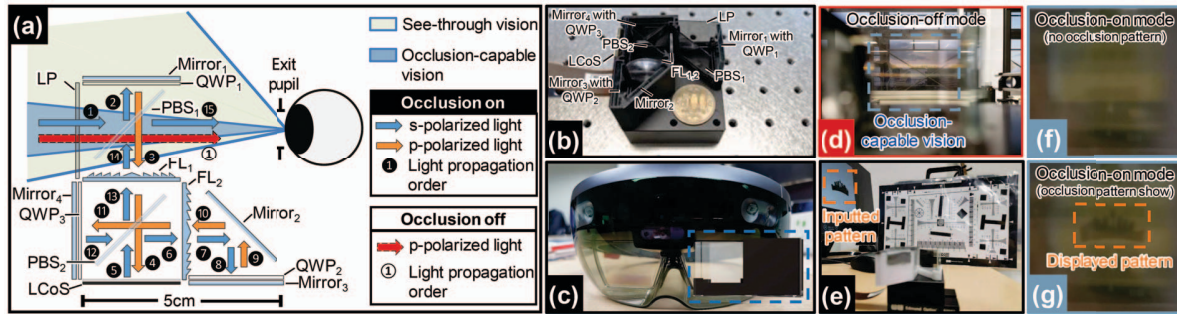


Figure 1: (a) Schematic diagram (top view) of the proposed occlusion module. The system optionally works under the occlusion-on mode or the occlusion-off mode by setting the polarization of the incident light with a linear polarizer (LP). (b) A prototype is built with a 3D-printed housing, and a coin ($\phi = 26.5\text{mm}$) is placed beside. (c) A conceptual diagram of the occlusion module being attached to HoloLens 1. (d) A see-through view captured under the occlusion-off mode. The central occlusion-capable vision is bounded by blue dash lines. (e) A preliminary experiment is conducted to test the prototype performance. The prototype is placed before a test chart and switched to the occlusion-on mode. A pattern shown on the monitor is inputted to the LCoS. (f) A blurred image of the front chart within the occlusion-capable vision is observed through the prototype, and (g) a sharp occlusion pattern is displayed.

ABSTRACT

The occlusion function benefits augmented reality (AR) in many aspects. However, existing occlusion-capable optical see-through augmented reality (OC-OST-AR) displays are designed by integrating virtual displays into a dedicated occlusion-capable architecture, hereby, we miss merits from emerging OST-AR displays. In this article, we propose an external occlusion module that can be added to common OST-AR displays. Per-pixel occlusion is supported with a small form-factor by using polarization-based optical path compression. The occlusion function can be switched on/off by controlling the incident light polarization. A prototype within a volume of $6 \times 6 \times 3\text{cm}$ is built. A preliminary experiment proves that occlusion is realized successfully.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Hardware—Emerging technologies—Emerging optical and photonic technologies;

1 INTRODUCTION

Augmented reality (AR) has been actively developed in recent years, as it is considered as a next-generation display technology that combines the physical world with digital information seamlessly. Optical see-through augmented reality (OST-AR) displays are one of the essential platforms to support daily used AR experiences. Digital contents in an OST-AR display are projected by either a laser source

or a display panel, then integrated into the real scene by an optical combiner before entering the users' eyes. In the presence of high ambient illumination, the projected virtual image faces problems including low visibility, weak perceptual cues, narrow color gamut, etc [1, 7]. In order to tackle those issues, occlusion-capable optical see-through augmented reality (OC-OST-AR) displays have been proposed and demonstrated to render per-pixel occlusion on the real scene that attenuates the background illumination arbitrarily. Following the initial system introduced by Kiyokawa et al. [3], various features, such as vari-focal occlusion [2] and a wide field-of-view (FOV) [8], have been achieved in OC-OST-AR displays.

Nevertheless, the practical application of OC-OST-AR displays is hindered by the large form-factor. The state-of-the-art device demonstrated a head-up but face-cover volume, which is noticeably larger than typical OST-AR displays [7]. Further compressing the form-factor faces the challenge by the dedicated optical architecture (a lens group to generate an inverted real image for conducting occlusion and the following part to erect the output image) and the requirement of providing a see-through scene with little distortion. In addition, existing AR display technologies are often designed to work with monochromatic light (e.g., holographic display [5] and wave-guide display [6]), while OC-OST-AR displays are built to process natural light among the visible spectrum. Merging some mainstream AR display technologies into OC-OST-AR displays is highly challenging. What is more, a spatial light modulator (SLM) and additional optical elements make an OC-OST-AR display expensive, which may cause it less appealing in the civilian market than common OST-AR displays.

Therefore, we propose add-on occlusion in this article. Different from existing OC-OST-AR displays that integrate a virtual display and occlusion into an optical architecture, an external module is hung before the optical combiner of common OST-AR displays to enable the occlusion function independently. Consequently, occlusion turns to be compatible with all the OST-AR displays as an optional feature.

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2 SYSTEM LAYOUT AND PROTOTYPING

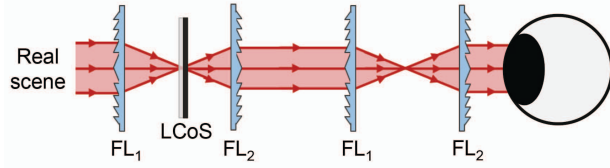


Figure 2: Unfolded structure of the proposed system.

An unfolded structure of the proposed system is shown as Fig. 2. Beams from the real scene are focused at the back focal plane of a Fresnel lens (FL), generating an inverted image for a Liquid Crystal on Silicon (LCoS) device to block the real scene on a pixel-by-pixel basis. Then, the processed real scene is erected and projected into the user's eye with a unit magnification by the following lenses. Notice that only a lens pair of FL₁ and FL₂ are implemented in this design. A typical occlusion-capable architecture is equivalently formed by making the incident light pass through each lens twice.

The schematic diagram (top view) of the proposed occlusion module is shown as Fig. 1 (a). Polarizing beam splitters (PBSs) and mirrors with quarter-wave plates (QWPs) are added to realize the reuse of the FL pair and compress the optical path. The see-through vision and occlusion-capable vision are drawn in green and blue, respectively. A linear polarizer (LP) is placed at the entrance pupil of the system to switch the system between occlusion on and off modes. Propagation light is drawn as blue and orange arrows in the occlusion-on mode and red arrows in the occlusion-off mode. Marks beside the arrows indicate the order of light propagation through each optical element in the two modes. When s-polarized light is generated by the LP (1), the PBS₁ reflects the incoming real scene to the mirror₁ (2). The QWP₁ is attached to the front surface of the mirror₁ so that it rotates the s-polarized light into p-polarized [7]. Hence, the reflected real scene by the mirror transmits the PBS₁ (3), then enters the following optical system for conducting occlusion through the FL₁. The PBS₂ is placed behind with a distance of half of the focal length of the FL₁. The p-polarized light passes through the PBS₂ (4), then reaches the LCoS fixed at the back focal plane of the FL₁. The non-occluded real scene is turned into s-polarized after being reflected backward on the LCoS surface (5) and guided to the FL₂ by the PBS₂ (6). The 45° tilted mirror₂ and the horizontal mirror₃ with the attached QWP₂ are used to guide the light backward with the polarization being switched into p direction (7)-(10). The spacing between the FL₂ and the mirror₃ is set to make the generated see-through image have the same depth as the original real scene for the eye at the exit pupil. Then, the reflected light propagates FL₂ and FL₁ again with polarization changed by the combination of the mirror₄ and the QWP₃ (11)-(13). Finally, the processed real scene is projected to the user's eye by the PBS₁ (14)-(15). Since the light passes through polarizing optical elements repeatedly in this design, the occlusion-capable vision is severely darkened, which may negatively impact the user experience in some poorly illuminated scenarios. Meanwhile, commercial OST-AR displays have been demonstrated to work well indoors. Hence, the occlusion function in this module is designed to be optionally turned off by rotating the LP by 90° (set the incident light to be p-polarized). Thus, the entire real scene would ignore the system and enter the user's pupil directly, as the red arrow illustrated in Fig. 1 (a).

A prototype is built as shown in Fig. 1 (b). Optical elements are mounted in a 3D-printed housing with a dimension of 6×6×3cm. The FL pair have the same focal length of 25mm and diameter of 25mm. The LP and the QWP have transmittances of 42% and 89%, respectively; the PBS has an average reflectance of 90% for the s-polarized light and transmittance of 90% for the p-polarized light;

the LCoS has a reflectance of 73%. As a result, the expected see-through efficiency of the prototype is calculated as 7.3%. Notice that optical elements in this early prototype have relatively low performance for minimizing the cost. The see-through efficiency can be as high as 21.2% when the PBS and QWP are replaced with more costly products.

A conceptual diagram of the prototype attached to HoloLens 1 is illustrated as Fig. 1 (c). The prototype is expected to be fixed closely before the wave-guide combiner (the front visor should be taken off previously). In order to make the exit pupils of the two devices overlap, the prototype is designed to form an exit pupil 50mm behind the LP according to the maximum eye relief of 25mm for typical AR headsets [4].

A preliminary experiment was conducted to examine the feasibility of rendering occlusion by the proposed system. The smartphone camera of HUAWEI P30 Pro is placed at the exit pupil to capture the output image. The captured see-through view under the occlusion-off mode is shown in Fig. 1 (d). Then, the prototype is switched to the occlusion-on mode and placed before a test chart, as shown in Fig. 1 (e). Fig. 1 (f) shows a blurred see-through view (cropped with the occlusion-capable vision) is captured by the camera due to the misalignment of FLs in the prototype. After an occlusion mask is inputted to the LCoS, a sharp occlusion pattern is displayed on the real background, as shown in Fig. 1 (g).

3 CONCLUSION

Mutual occlusion has been demonstrated to be a crucial feature for OST-AR displays, however, existing OC-OST-AR displays are still away from a practical solution. Hence, we propose add-on occlusion as a compatible solution in this article. The external occlusion module benefits typical OST-AR displays to conduct sharp occlusion without modifying imaging systems. A prototype is built, and the preliminary experiment proves the feasibility of rendering sharp occlusion with the proposed design. The prototype is expected to be optimized with better image quality and higher see-through efficiency. Then, experiments associated with commercial OST-AR displays and corresponding user studies can be conducted in the future.

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